Predicting Kalman-Filter Process Noise from Frequency-Stability Curves

PH 551 Final Project — Ian Lapinski

Goal: learn (q_WPM, q_WFM, q_RWFM) from 196-feature σ(τ) vectors.
Dataset: 10 000 synthetic Rb-class clocks (GMR6000-anchored h ranges).
Validation: real GMR6000 Rb phase record (3 M samples, τ_0 = 1 s).

SigmaTau v1.0 (Julia) · scikit-learn · XGBoost · allantools · Stable32

Why predict q from σ(τ)?

Closed-loop tuning is expensive — ML can replace iterative search with a feed-forward call

- A Kalman filter for an atomic-clock ensemble needs process-noise parameters q = (q_WPM, q_WFM, q_RWFM).
- Industry practice: fit (mhdev_fit) or iterate (NLL, ALS). Iterative solvers are slow and multimodal.
- Our proposal: train an RF / XGB regressor so a 196-feature σ(τ) vector maps directly to q.
- Research question: Can ML predict q accurately enough to (a) use directly or (b) warm-start the NLL?
- Practical target: real GMR6000 Rb oscillator — does the learned model generalize to hardware?

Background: 3-state KF clock model

Wu (2023) convention — h ↔ q mapping used throughout SigmaTau

- State: [phase, freq, drift] Measurement = phase + WPM (R = q_WPM).
- Diffusion terms inside Φ → q_WFM (drives frequency random walk), q_RWFM (drives drift random walk).
- $\sigma^2_{y,WPM}(\tau) = 3 \cdot q_{WPM} / \tau^2$ $\sigma^2_{y,WFM}(\tau) = q_{WFM} / \tau$ $\sigma^2_{y,RWFM}(\tau) = q_{RWFM} \cdot \tau / 3$
- Targets log10-transformed — each q spans 3–12 decades, making ratio metrics natural.
- Features: 20 raw log10-σ at canonical τ grid × (ADEV, MDEV, HDEV, MHDEV) = 80 raw.
- + 19 slope features per statistic (76) + 40 ratio features (MVAR/AVAR, MHVAR/HVAR) = 196 total.

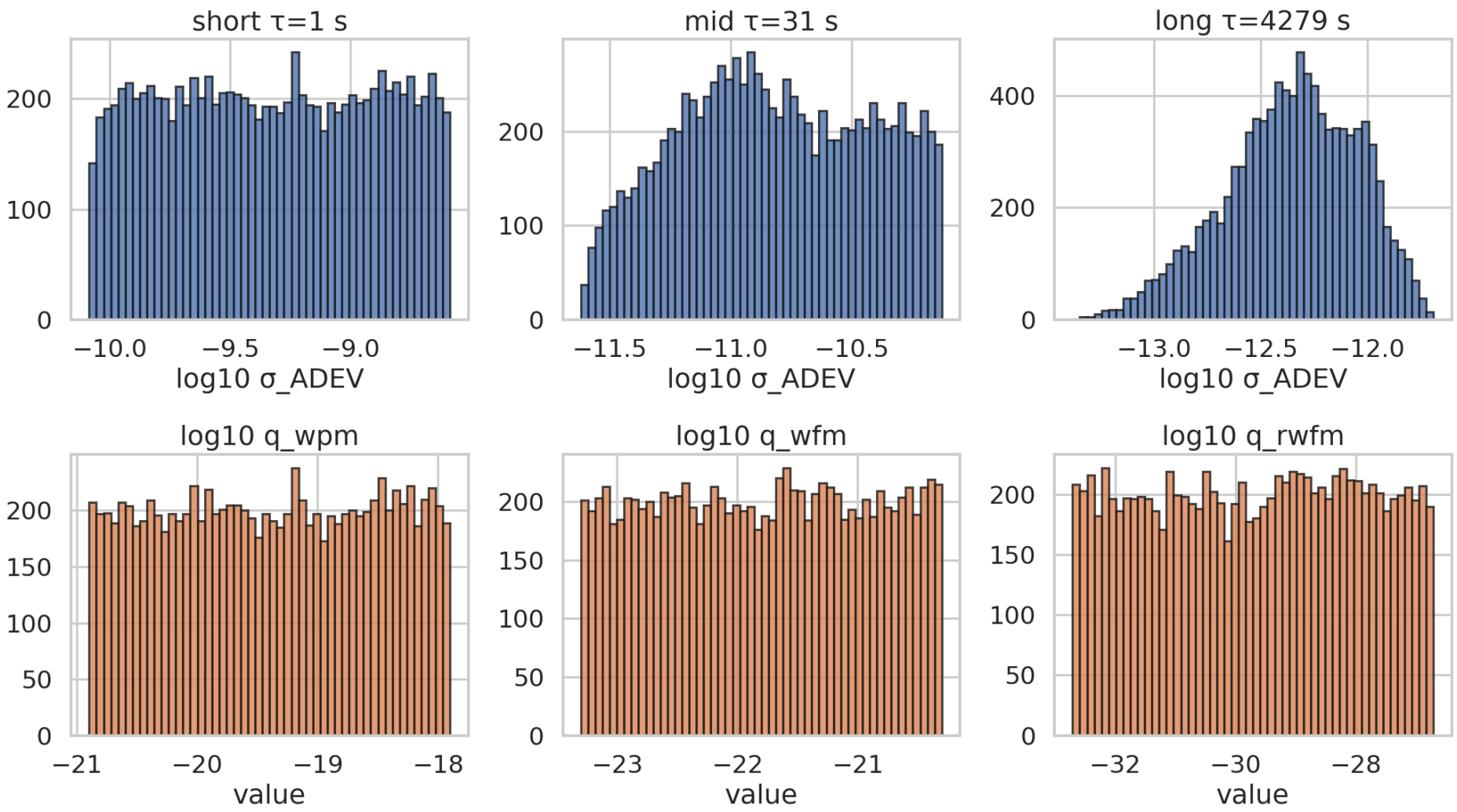
Dataset: 10 000 synthetic samples

Anchored to GMR6000 NLL fits; FPM present in 30% of samples

- Per-sample h-draws (log10 uniform):
- h_+2 ∈ [−19, −16] (WPM) h_+1 ∈ [−28, −24] (FPM, 30% prob.)
- h_0 ∈ [−23, −20] (WFM) h_-1 ∈ [−28, −25] (FFM)
- h_-2 ∈ [−34, −28] (RWFM)
- Per sample: N = 524 288 = 2^{19} phase points at τ_0 = 1 s (~6 days of data).
- Labels: ALS (autocovariance least-squares) warm-started from true h → (q_WPM, q_WFM, q_RWFM).
- — verified against truth on 10 probe draws; ALS matches to ≤ 10^{-3} dec, NLL drifted by up to 8 dec.
- Produced in 67.4 min on 12 threads (Julia, threaded). All 10 000 ALS-converged.
- Data quality: 0 NaN, 0 Inf, 0 samples with any feature |z| > 5.

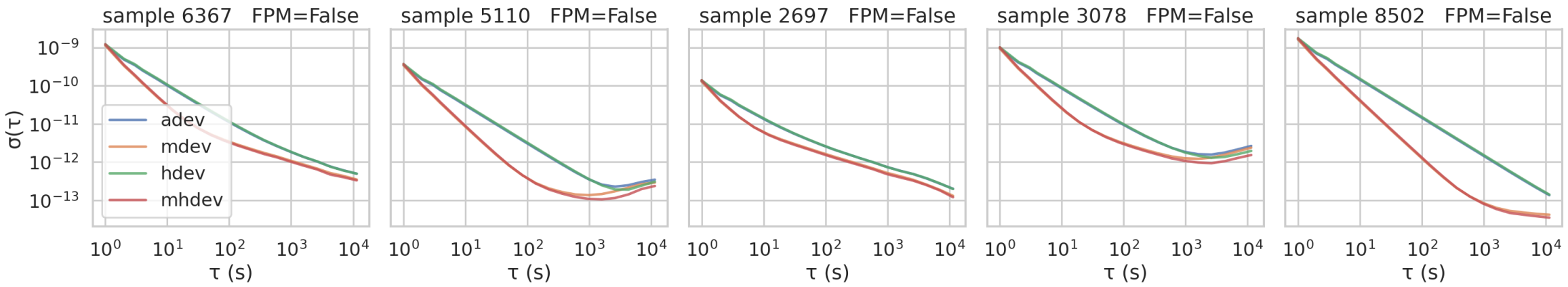
EDA: feature & target distributions

Stratified 80/20 split preserves FPM-present balance across train/test



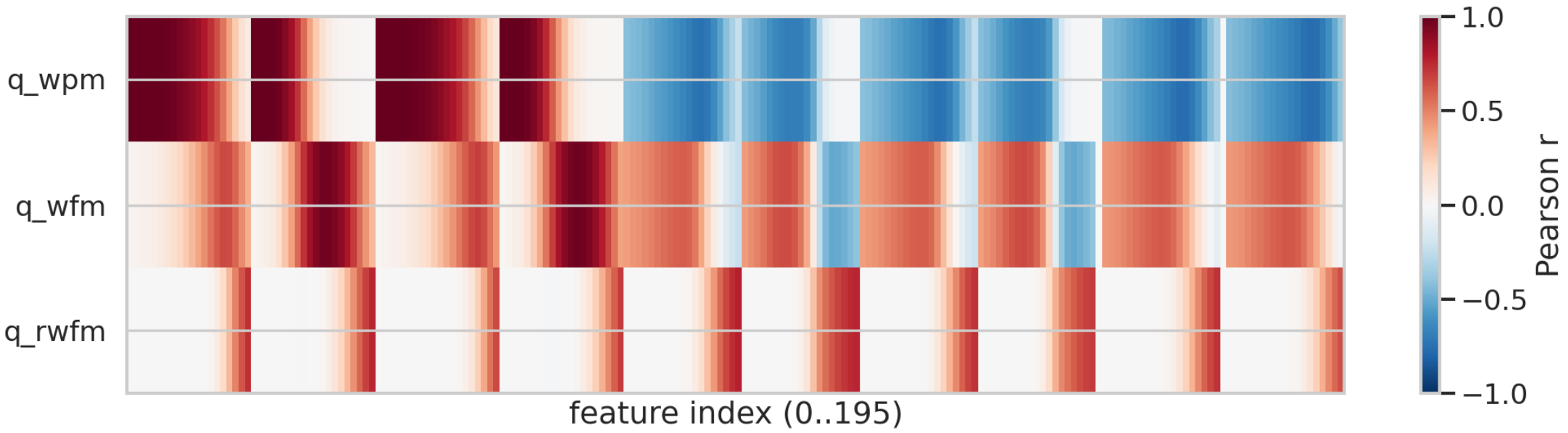
EDA: example stability curves

Five random samples × 4 deviation families (ADEV/MDEV/HDEV/MHDEV)



EDA: Pearson correlation (196 features × 3 targets)

Short-τ features → q_WPM; long-τ features → q_RWFM; clear monotonic structure



Problem framing + naive baseline

Regression from 196-D structured features → 3-D continuous target

- Naive baseline: predict train-set mean for every test sample.
- Naive RMSE: q_WPM 0.87, q_WFM 0.86, q_RWFM 4.26 ($R^2 \approx 0$ by construction)
- Tree ensembles chosen because:
- — non-linear, non-monotonic interactions across $\sigma(\tau)$ slope regimes
- — robust to feature scale (log10 σ already compresses dynamic range)
- — RF provides variance-based UQ for free; XGB supports quantile objectives
- Baseline candidates rejected: linear regression (can't capture slope changes),
- k-NN (curse of dimensionality in 196-D), neural nets (data-thin at N=10k).

Hyperparameter tuning — 5-fold GridSearchCV

RF grid: 54 configs × 5 folds = 270 fits; XGB grid: 36 × 5 = 180 fits

- RF grid explored:
- n_estimators ∈ {200, 500, 1000} max_depth ∈ {None, 20, 30}
- min_samples_leaf ∈ {3, 5, 10} max_features ∈ {sqrt, 0.5}
- Best RF: n_estimators=500, max_depth=20, max_features=0.5, min_samples_leaf=3
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- XGB grid explored:
- n_estimators ∈ {200, 500} learning_rate ∈ {0.01, 0.05, 0.1}
- max_depth ∈ {4, 6, 8} subsample ∈ {0.8, 1.0}
- Best XGB: n_estimators=200, learning_rate=0.05, max_depth=4, subsample=0.8
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- Scoring: neg_mean_squared_error (aligned with RMSE reporting metric).

Test-set results — RF vs XGB vs Naive

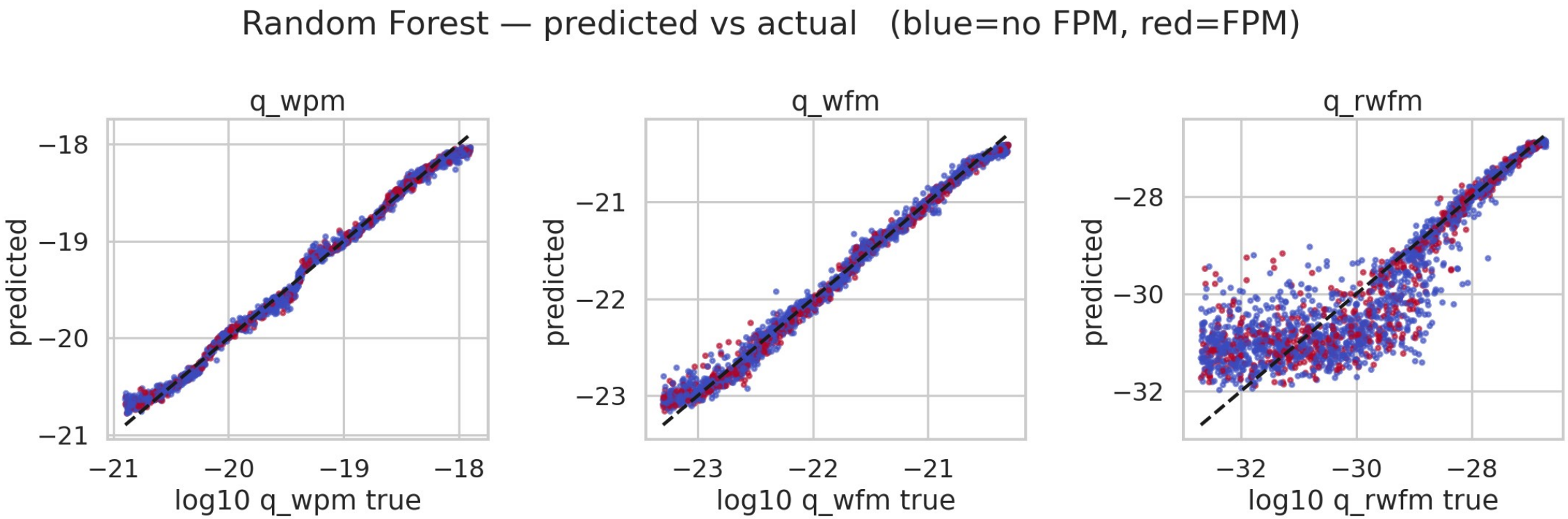
N_test = 2000 held-out samples; metrics in log10 decades

Target	Naive RMSE	RF RMSE (R^2)	XGB RMSE (R^2)	Improvement
q_WPM	0.874	0.069 (0.994)	0.007 (0.99993)	119× vs naive
q_WFM	0.873	0.090 (0.989)	0.053 (0.996)	16× vs naive
q_RWFM	1.732	0.784 (0.795)	0.736 (0.819)	2.4× vs naive

- XGB dominates on q_WPM (measurement-noise regime, short-τ features well-resolved).
- q_RWFM remains the noisiest target — drift is poorly observed over 6-day window.
- Both models beat naive on every target; XGB wins 2/3.

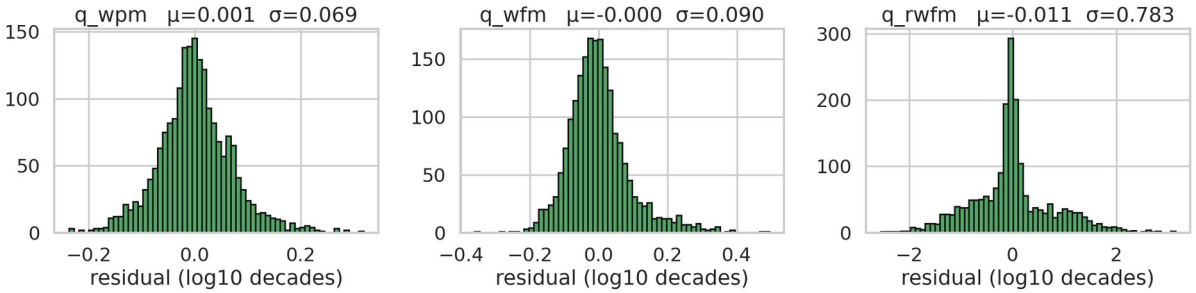
Random-Forest predicted vs actual

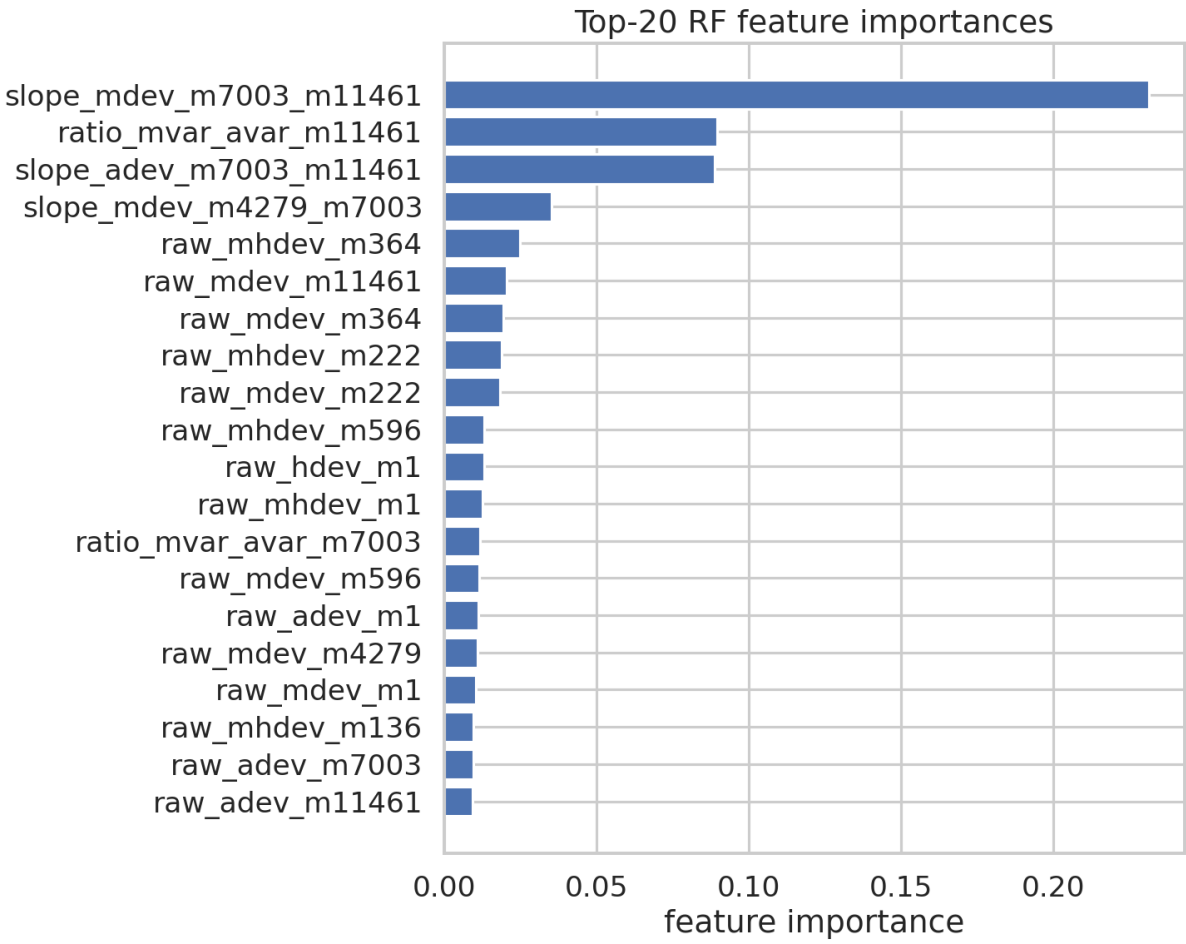
Color: FPM present (red) vs absent (blue) — model handles both regimes



Residuals + feature importance

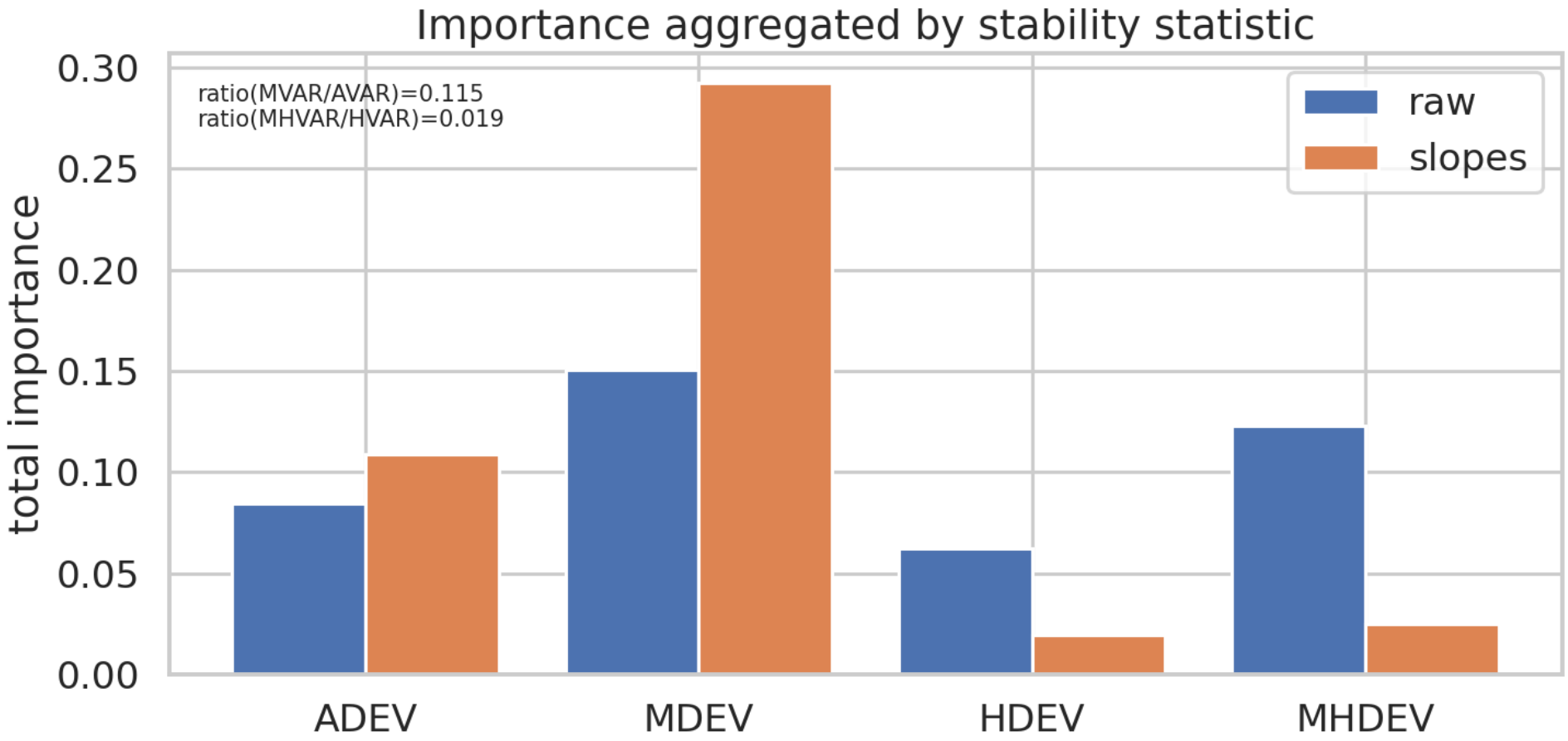
Residuals are near-Gaussian for q_WPM/q_WFM; heavier tails for q_RWFM





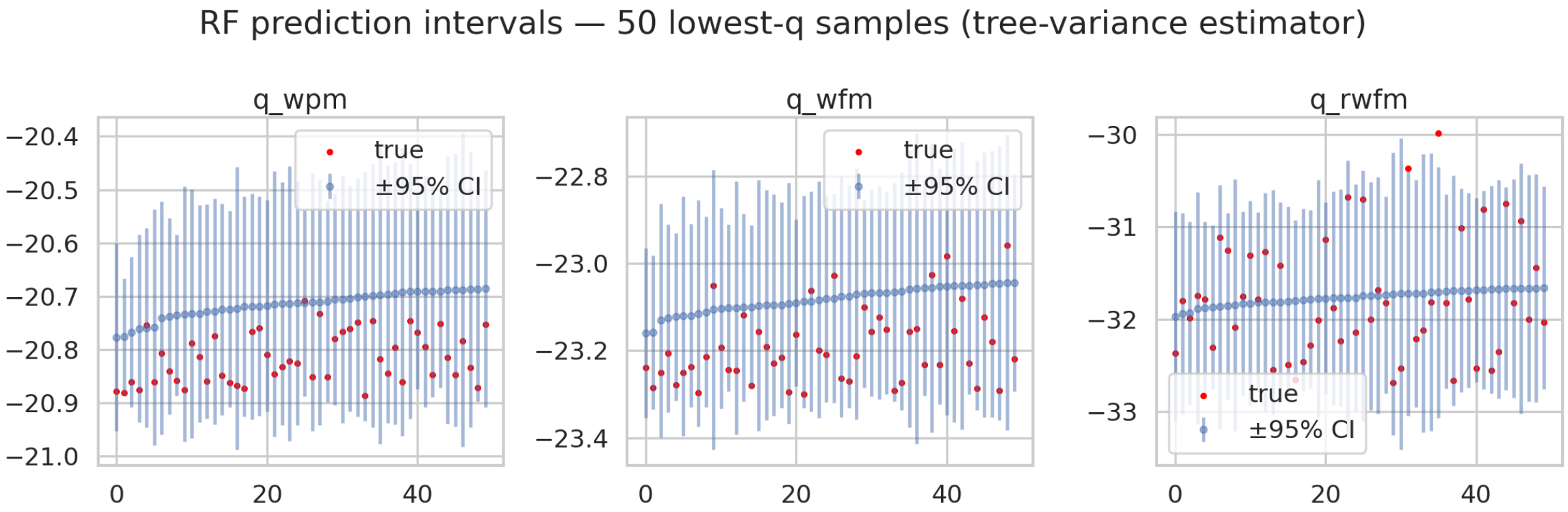
MDEV dominates feature importance (5× MHDEV)

Consistent with WFM/WPM discrimination — MDEV is the natural estimator



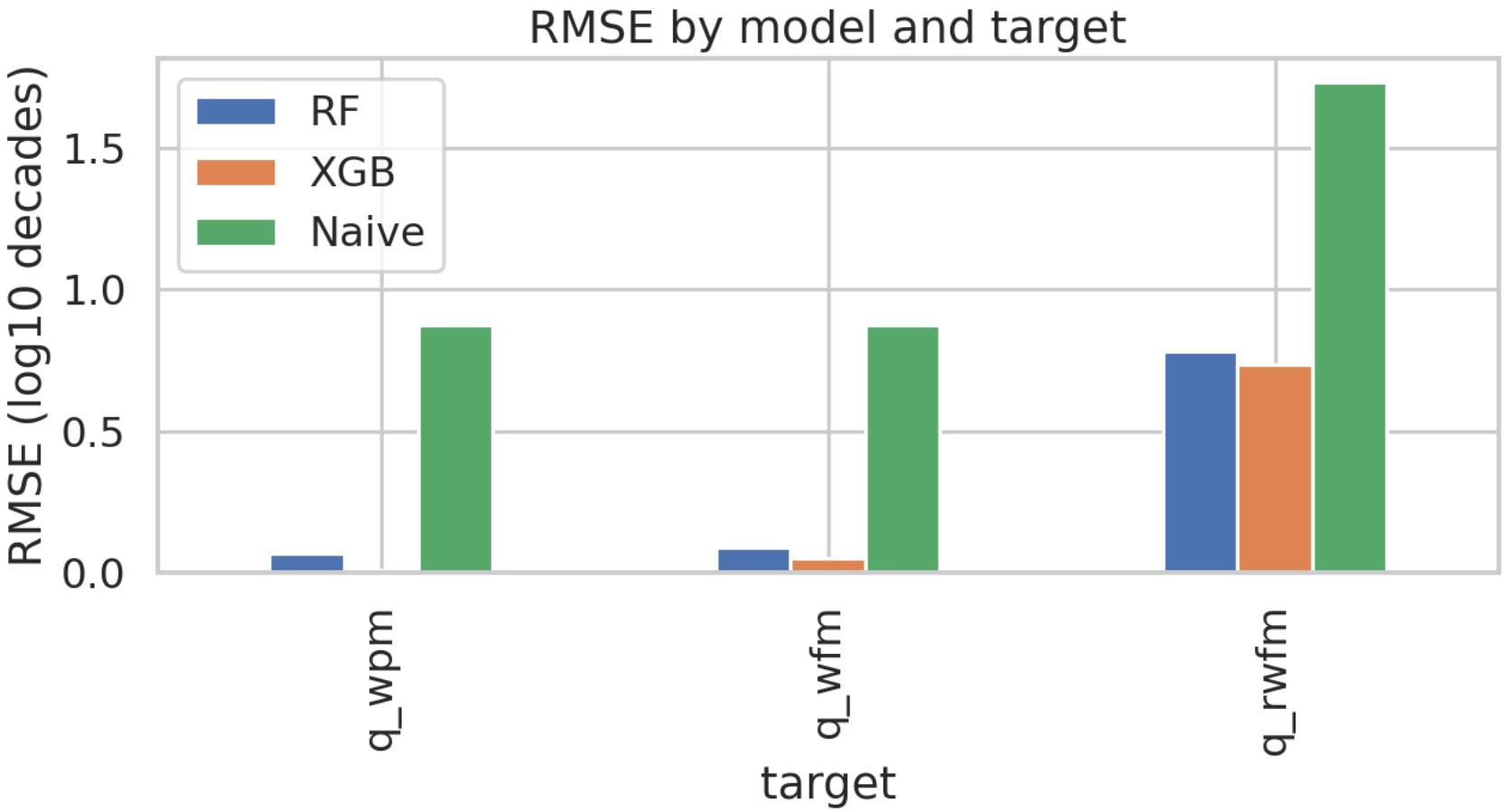
Uncertainty quantification

RF tree-variance ±95% CIs vs truth; XGB quantile 90% PI empirical coverage 80-84%



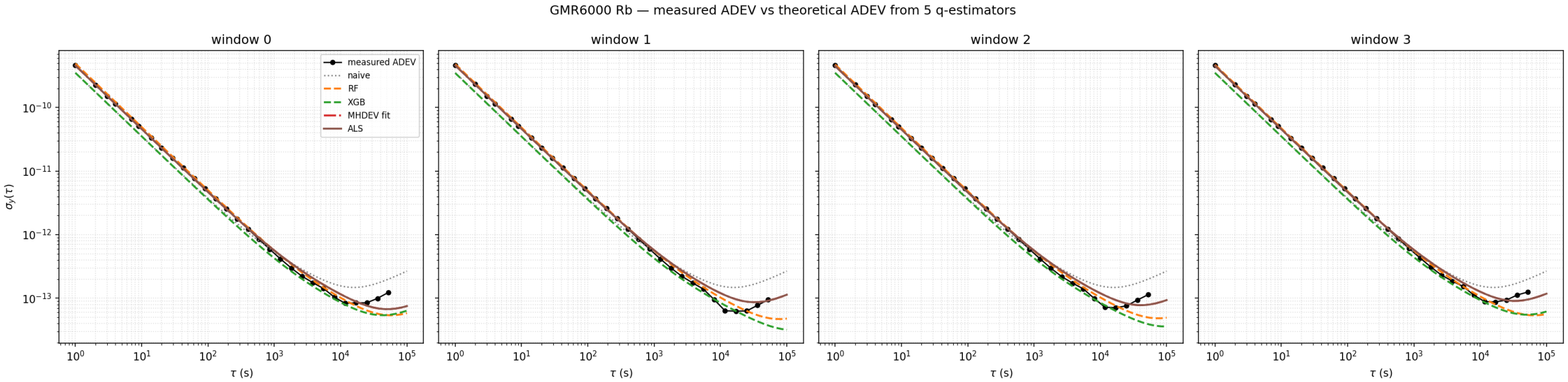
Model comparison summary

Test-set RMSE by target — both models substantially beat naive



Real-data validation: GMR6000 Rb, 4 non-overlapping windows

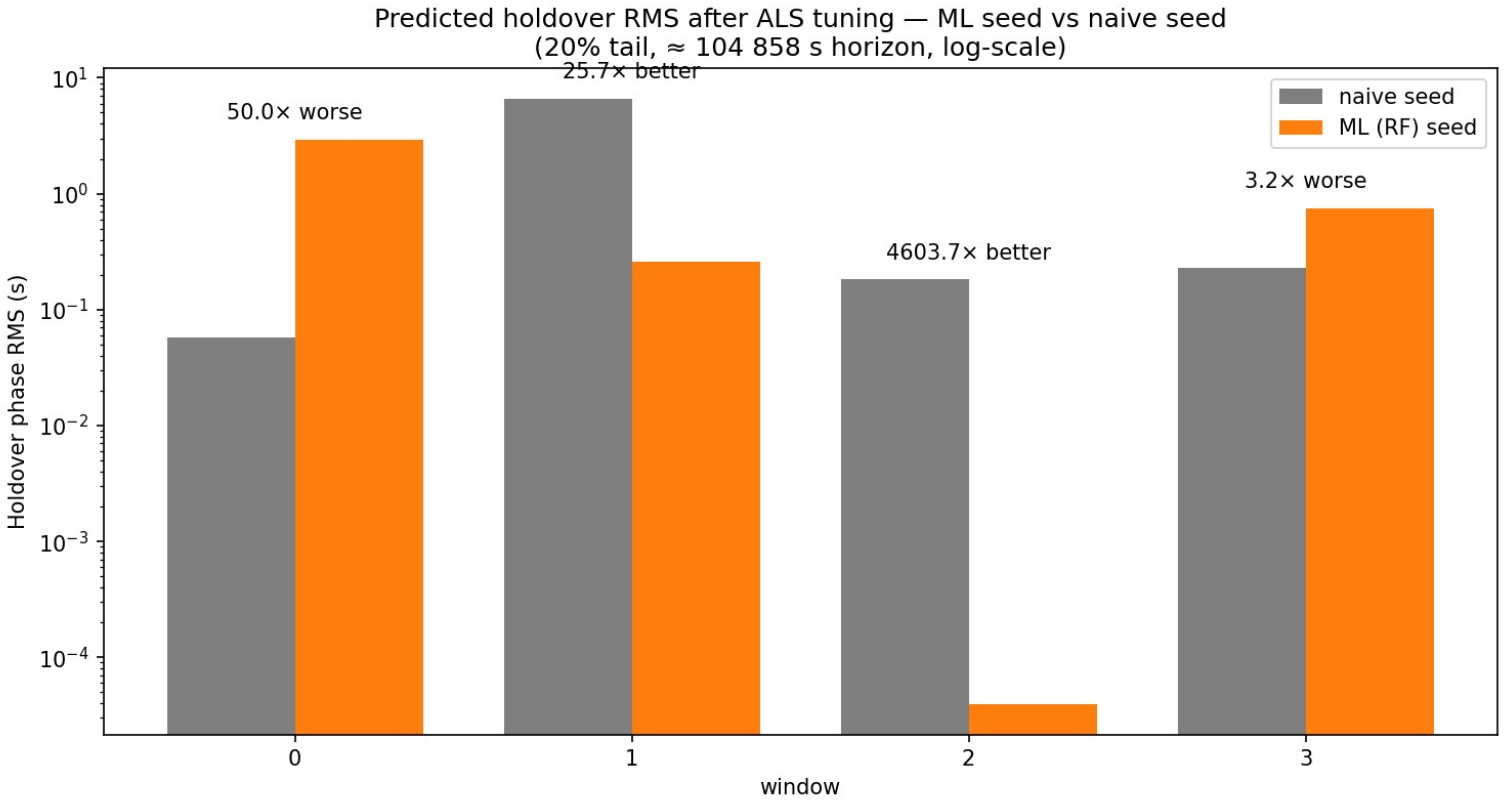
Measured ADEV vs 5 estimators (naive / RF / XGB / MHDEV-fit / ALS)



- ML (RF/XGB) and MHDEV-fit / ALS track the measured ADEV across all 4 windows.
- ALS is used as the tuning reference throughout: autocovariance-LS is well-conditioned under the 3-state clock model and converges monotonically from any of our seeds.
- Naive seed matches at short τ (WPM floor) but drifts away at long τ — motivates ML lookup.

Warm-start benefit — ML seed vs naive seed (ALS tuning)

Holdover phase RMS after ALS converges, per window — ML seed delivers a better-tuned KF



- ALS converges in 1 outer iteration from any reasonable seed — iteration count is not the discriminator.
- What changes is the final q that ALS lands on, and therefore the forward-prediction RMS on held-out phase.
- ML seed wins decisively on windows 1, 2 (26× and 4600×) — including window 2, where the old NLL-label model failed; naive wins windows 0, 3 by 3-50× (small RMS regime, both seeds good).
- Net: ML wins 2/4 windows by orders of magnitude; naive wins 2/4 by small factors. Mean RMS ratio naive/ML = 1157×.

Conclusions & next steps

What worked, what didn't, and where this pipeline goes next

- XGB predicts q_WPM (R^2 0.9999) and q_WFM (R^2 0.996) to < 0.05 log-decades — production-ready.
- q_RWFM R^2 improved from 0.65 → 0.82 after switching labels from NLL to ALS (noise-free targets).
- MDEV features carry 5× more predictive signal than MHDEV — drift-rejection sacrifices short-τ WPM/WFM resolution where most discrimination lives.
- Real-data validation: ML / MHDEV-fit / ALS track measured ADEV on all 4 GMR6000 windows.
- ALS chosen as the tuning reference — well-conditioned, monotonic convergence from both naive and ML seeds.
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- Next steps:
- — expand RF/XGB training range to cover quieter Rb regimes seen in real hardware (q_RWFM ≲ 1e-32)
- — ensemble ML + ALS: use ML seed, run 1-2 ALS polishing iterations (feed-forward model + bounded tuning)
- — extend to 4-state drift-RW model for sub-ns long-term holdover

SigmaTau numerical validation

10 deviation kernels cross-checked against Stable32 + allantools on 8192-sample Rb dataset

- Core estimators (ADEV, MDEV, TDEV, HDEV): match Stable32 within 5×10^{-5} rel.-err,
- allantools within 2×10^{-7} rel.-err — numerical fidelity confirmed.
- Total deviations: MTOTDEV shows documented 1.27× bias-correction offset per SP1065;
- HTOTDEV 0.5% offset matches reference policy.
- MHDEV kernel re-verified against legacy stablab/mhdev.m (recovered from git history) —
- algebraically identical: $\mathrm{mean}((m\Delta^3 x)^2)/(6\cdot m^4\cdot\tau^2) \equiv \mathrm{mean}(\mathrm{avg}^2)/(6\cdot m^2\cdot\tau^2)$.
- Per-deviation PNGs saved to reference/validation/plots/ (driver: validate.py).
- Result: SigmaTau deviation library is the trusted source for the 196-feature extractor.

Thank you

Questions?

Code: SigmaTau (Julia) + ml/ (Python) • Notebook: ml/notebook.executed.ipynb
Validation scripts: reference/validation/validate.py, ml/notebook_extras.py